

Riphah International University

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3rd Semester

Computer Organization & AL

Lab Task 10:

Code 1(Bed):

```
.MODEL SMALL
```

```
.STACK 100H
```

```
.DATA
```

```
    BED_COLOR EQU 14
```

```
.CODE
```

```
PUT_PIXEL MACRO
```

```
    MOV AH, 0CH
```

```
    MOV AL, BED_COLOR
```

```
    INT 10H
```

```
ENDM
```

```
MAIN PROC
```

```
    MOV AX, @DATA
```

```
    MOV DS, AX
```

```
    MOV AX, 12H
```

```
    INT 10H
```

```
    CALL DRAW_HEADBOARD
```

```
    CALL DRAW_MATTRESS
```

```
    CALL DRAW_LEGS
```

```
    MOV AH, 00H
```

```
    INT 16H
```

```
    MOV AX, 03H
```

```
    INT 10H
```

```
    MOV AH, 4CH
```

INT 21H

MAIN ENDP

DRAW_HEADBOARD PROC

MOV SI, 100

MOV DI, 220

MOV BX, 320

MOV BP, 440

MOV CX, BX

MOV DX, SI

HL_TOP:

PUT_PIXEL

INC CX

CMP CX, BP

JLE HL_TOP

MOV CX, BX

MOV DX, DI

HL_BOT:

PUT_PIXEL

INC CX

CMP CX, BP

JLE HL_BOT

MOV CX, BX

MOV DX, SI

HL_LEFT:

PUT_PIXEL

INC DX

CMP DX, DI

JLE HL_LEFT

MOV CX, BP

MOV DX, SI

HL_RIGHT:

PUT_PIXEL

INC DX

CMP DX, DI

JLE HL_RIGHT

MOV CX, 380

MOV DX, 100

HL_VCENTER:

PUT_PIXEL

INC DX

CMP DX, 220

JLE HL_VCENTER

MOV CX, 320

MOV DX, 160

HL_HCENTER:

PUT_PIXEL

INC CX

CMP CX, 440

JLE HL_HCENTER

RET

DRAW_HEADBOARD ENDP

DRAW_MATTRESS PROC

MOV CX, 320

MOV DX, 220

MAT_L_DIAG:

PUT_PIXEL

DEC CX

INC DX

CMP DX, 320

JLE MAT_L_DIAG

MOV CX, 440

MOV DX, 220

MAT_R_DIAG:

PUT_PIXEL

DEC CX

INC DX

CMP DX, 320

JLE MAT_R_DIAG

MOV CX, 220

MOV DX, 320

MAT_FRONT:

PUT_PIXEL

INC CX

CMP CX, 340

JLE MAT_FRONT

MOV CX, 350

MOV DX, 220

MAT_PAT_1:

PUT_PIXEL

DEC CX

INC DX

CMP DX, 320

JLE MAT_PAT_1

MOV CX, 380

MOV DX, 220

MAT_PAT_2:

PUT_PIXEL

DEC CX

INC DX

CMP DX, 320

JLE MAT_PAT_2

RET

DRAW_MATTRESS ENDP

DRAW_LEGS PROC

MOV CX, 220

MOV DX, 320

LEG_FL_V:

PUT_PIXEL

INC DX

CMP DX, 400

JLE LEG_FL_V

MOV CX, 215

MOV DX, 400

LEG_FL_F:

PUT_PIXEL

INC CX

CMP CX, 225

JLE LEG_FL_F

MOV CX, 340

MOV DX, 320

LEG_FR_V:

PUT_PIXEL

INC DX

CMP DX, 400

JLE LEG_FR_V

MOV CX, 335

MOV DX, 400

LEG_FR_F:

PUT_PIXEL

INC CX

CMP CX, 345

JLE LEG_FR_F

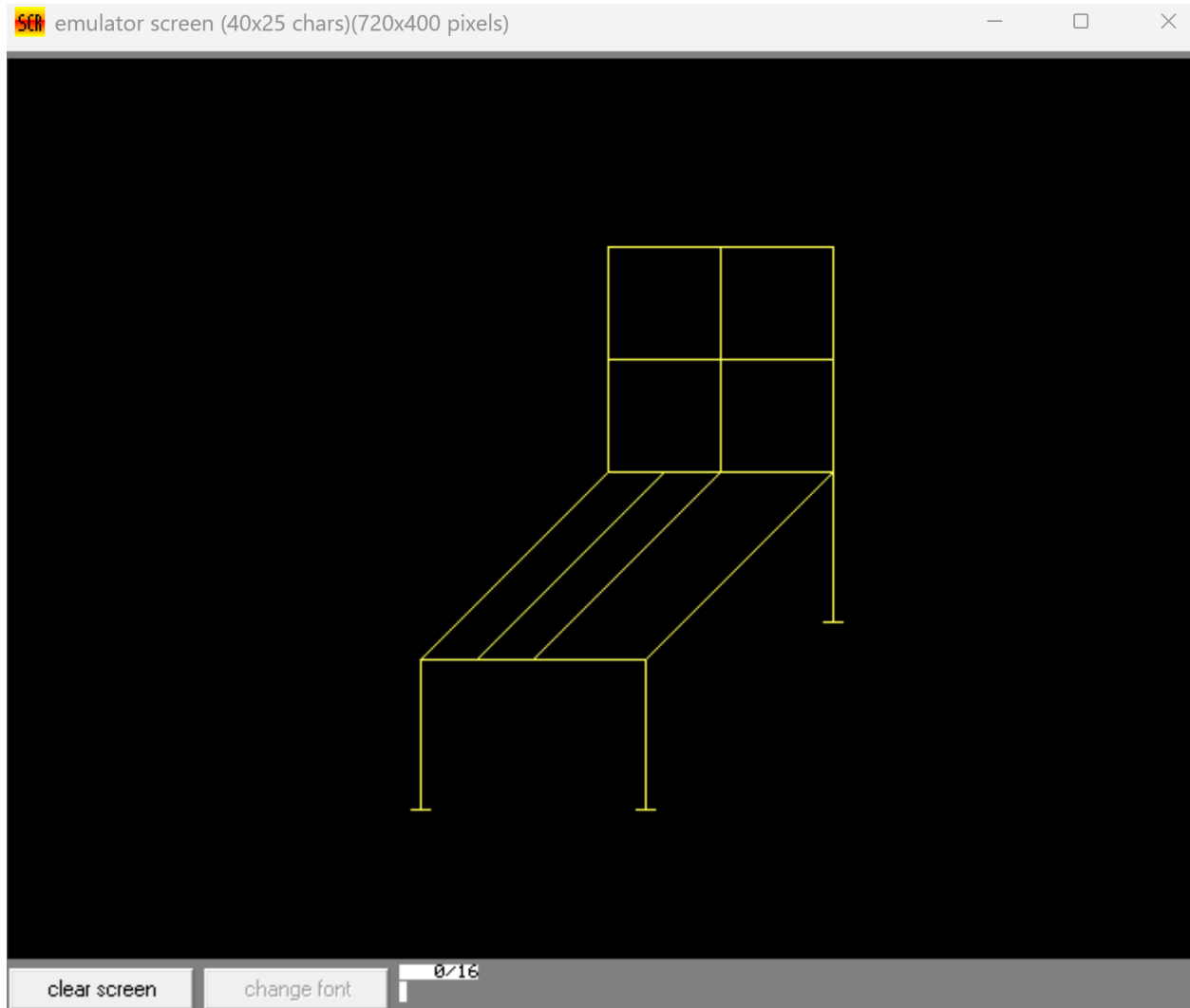
MOV CX, 440

MOV DX, 220

LEG_BR_V:

```
    PUT_PIXEL
    INC DX
    CMP DX, 300
    JLE LEG_BR_V
    MOV CX, 435
    MOV DX, 300
LEG_BR_F:
    PUT_PIXEL
    INC CX
    CMP CX, 445
    JLE LEG_BR_F
    RET
DRAW_LEGS ENDP
END MAIN
```


Output:



Code Description:

Purpose: The program renders a stylized graphical diagram of a bed using 8086 Assembly language in **VGA Graphics Mode 12h** (640x480 resolution).

Key Features:

- **Graphics Setup:** It switches the video adapter to graphics mode to allow pixel-by-pixel drawing.
- **Macro Usage:** It defines a PUT_PIXEL macro that utilizes Interrupt 10h to simplify the process of plotting individual points on the screen.
- **Modular Design:** The drawing logic is split into three distinct procedures:

1. DRAW_HEADBOARD: Draws a rectangular frame with a decorative **cross pattern** in the center.
 2. DRAW_MATTRESS: Renders the bed surface in perspective with **diagonal pattern lines**.
 3. DRAW_LEGS: Draws vertical legs that now include specific "**feet**" at the bottom for detail.
- **Execution Flow:** The program draws the image in **Yellow** (Color 14), waits for a user keypress, and then gracefully returns the system to text mode before exiting.

Code 2(Table):

```
.MODEL SMALL
```

```
.STACK 100H
```

```
.DATA
```

```
TABLE_COLOR DB 6
```

```
DETAIL_COLOR DB 4
```

```
FOOT_COLOR DB 14
```

```
.CODE
```

```
MAIN PROC
```

```
MOV AX, @DATA
```

```
MOV DS, AX
```

```
MOV AL, 13H
```

```
MOV AH, 0
```

```
INT 10H
```

```
MOV CX, 120
```

```
MOV DX, 10
```

```
MOV AH, 0CH
```

MOV AL, TABLE_COLOR

MOV BL, 120

OUTER_1:

INT 10H

INC CX

DEC BL

JNZ OUTER_1

MOV AL, DETAIL_COLOR

MOV BL, 100

OUTER_2:

INT 10H

INC DX

DEC CX

DEC BL

JNZ OUTER_2

MOV AL, TABLE_COLOR

MOV BL, 120

OUTER_3:

INT 10H

DEC CX

DEC BL

JNZ OUTER_3

MOV AL, DETAIL_COLOR

MOV BL, 100

OUTER_4:

INT 10H

DEC DX

INC CX

DEC BL

JNZ OUTER_4

MOV CX, 240

MOV DX, 10

MOV AL, FOOT_COLOR

MOV BL, 8

INNER_1:

INT 10H

INC CX

DEC BL

JNZ INNER_1

MOV AL, DETAIL_COLOR

MOV BL, 110

INNER_2:

INT 10H

INC DX

DEC CX

DEC BL

JNZ INNER_2

MOV AL, FOOT_COLOR

MOV BL, 118

INNER_3:

INT 10H

DEC CX

DEC BL

JNZ INNER_3

MOV AL, DETAIL_COLOR

MOV BL, 8

INNER_4:

INT 10H

DEC DX

DEC BL

JNZ INNER_4

MOV CX, 20

MOV DX, 112

MOV AL, TABLE_COLOR

MOV BL, 50

LEG1_1:

INT 10H

INC DX

DEC BL

JNZ LEG1_1

MOV AL, DETAIL_COLOR

MOV BL, 7

LEG1_2:

INT 10H

INC DX

INC CX

DEC BL

JNZ LEG1_2

MOV AL, DETAIL_COLOR

MOV BL, 7

LEG1_3:

INT 10H

DEC DX

INC CX

DEC BL

JNZ LEG1_3

MOV AL, TABLE_COLOR

MOV BL, 44

LEG1_4:

INT 10H

DEC DX

DEC BL

JNZ LEG1_4

PUSH CX

PUSH DX

SUB CX, 14

ADD DX, 50

MOV BL, 15

F_LOOP1:

MOV AL, FOOT_COLOR

INT 10H

INC CX

DEC BL

JNZ F_LOOP1

POP DX

POP CX

MOV CX, 236

MOV DX, 23

MOV AL, TABLE_COLOR

MOV BL, 55

LEG2_1:

INT 10H

INC DX

DEC BL

JNZ LEG2_1

MOV AL, DETAIL_COLOR

MOV BL, 7

LEG2_2:

INT 10H

INC DX

INC CX

DEC BL

JNZ LEG2_2

MOV AL, DETAIL_COLOR

MOV BL, 7

LEG2_3:

INT 10H

DEC DX

INC CX

DEC BL

JNZ LEG2_3

MOV AL, TABLE_COLOR

MOV BL, 66

LEG2_4:

INT 10H

DEC DX

DEC BL

JNZ LEG2_4

PUSH CX

PUSH DX

MOV CX, 236

ADD DX, 55

MOV BL, 15

F_LOOP2:

MOV AL, FOOT_COLOR

INT 10H

INC CX

DEC BL

JNZ F_LOOP2

POP DX

POP CX

MOV CX, 140

MOV DX, 110

MOV AL, TABLE_COLOR

MOV BL, 55

LEG3_1:

INT 10H

INC DX

DEC BL

JNZ LEG3_1

MOV AL, DETAIL_COLOR

MOV BL, 7

LEG3_2:

INT 10H

INC DX

INC CX

DEC BL

JNZ LEG3_2

MOV AL, DETAIL_COLOR

MOV BL, 7

LEG3_3:

INT 10H

DEC DX

INC CX

DEC BL

JNZ LEG3_3

MOV AL, TABLE_COLOR

MOV BL, 62

LEG3_4:

INT 10H

DEC DX

DEC BL

JNZ LEG3_4

PUSH CX

PUSH DX

MOV CX, 140

ADD DX, 165

MOV BL, 15

F_LOOP3:

MOV AL, FOOT_COLOR

INT 10H

INC CX

DEC BL

JNZ F_LOOP3

POP DX

POP CX

MOV CX, 75

MOV DX, 120

MOV AL, TABLE_COLOR

MOV BL, 45

LEG4_1:

INT 10H

INC DX

DEC BL

JNZ LEG4_1

MOV AL, DETAIL_COLOR

MOV BL, 7

LEG4_2:

INT 10H

INC DX

INC CX

DEC BL

JNZ LEG4_2

MOV AL, DETAIL_COLOR

MOV BL, 7

LEG4_3:

INT 10H

DEC DX

INC CX

DEC BL

JNZ LEG4_3

MOV AL, TABLE_COLOR

MOV BL, 45

LEG4_4:

INT 10H

DEC DX

DEC BL

JNZ LEG4_4

PUSH CX

PUSH DX

MOV CX, 75

ADD DX, 165

MOV BL, 15

F_LOOP4:

MOV AL, FOOT_COLOR

INT 10H

INC CX

DEC BL

JNZ F_LOOP4

POP DX

POP CX

MOV CX, 130

MOV DX, 25

MOV AL, TABLE_COLOR

MOV BL, 75

CONC_1A:

INT 10H

INC CX

DEC BL

JNZ CONC_1A

MOV AL, DETAIL_COLOR

MOV BL, 75

CONC_1B:

INT 10H

INC DX

DEC CX

DEC BL

JNZ CONC_1B

MOV AL, FOOT_COLOR

MOV BL, 75

CONC_1C:

INT 10H

DEC CX

DEC BL

JNZ CONC_1C

MOV AL, DETAIL_COLOR

MOV BL, 75

CONC_1D:

INT 10H

DEC DX

INC CX

DEC BL

JNZ CONC_1D

MOV CX, 140

MOV DX, 40

MOV AL, TABLE_COLOR

MOV BL, 40

CONC_2A:

INT 10H

INC CX

DEC BL

JNZ CONC_2A

MOV AL, DETAIL_COLOR

MOV BL, 50

CONC_2B:

INT 10H

INC DX

DEC CX

DEC BL

JNZ CONC_2B

MOV AL, FOOT_COLOR

MOV BL, 40

CONC_2C:

INT 10H

DEC CX

DEC BL

JNZ CONC_2C

MOV AL, DETAIL_COLOR

MOV BL, 50

CONC_2D:

INT 10H

DEC DX

INC CX

DEC BL

JNZ CONC_2D

MOV CX, 145

MOV DX, 45

MOV AL, FOOT_COLOR

MOV BL, 20

CONC_3A:

INT 10H

INC CX

DEC BL

JNZ CONC_3A

MOV AL, DETAIL_COLOR

MOV BL, 30

CONC_3B:

INT 10H

INC DX

DEC CX

DEC BL

JNZ CONC_3B

MOV AL, FOOT_COLOR

MOV BL, 20

CONC_3C:

```
    INT 10H

    DEC CX

    DEC BL

    JNZ CONC_3C

    MOV AL, DETAIL_COLOR

    MOV BL, 30

CONC_3D:

    INT 10H

    DEC DX

    INC CX

    DEC BL

    JNZ CONC_3D

    MOV AH, 00H

    INT 16H

    MOV AX, 03H

    INT 10H

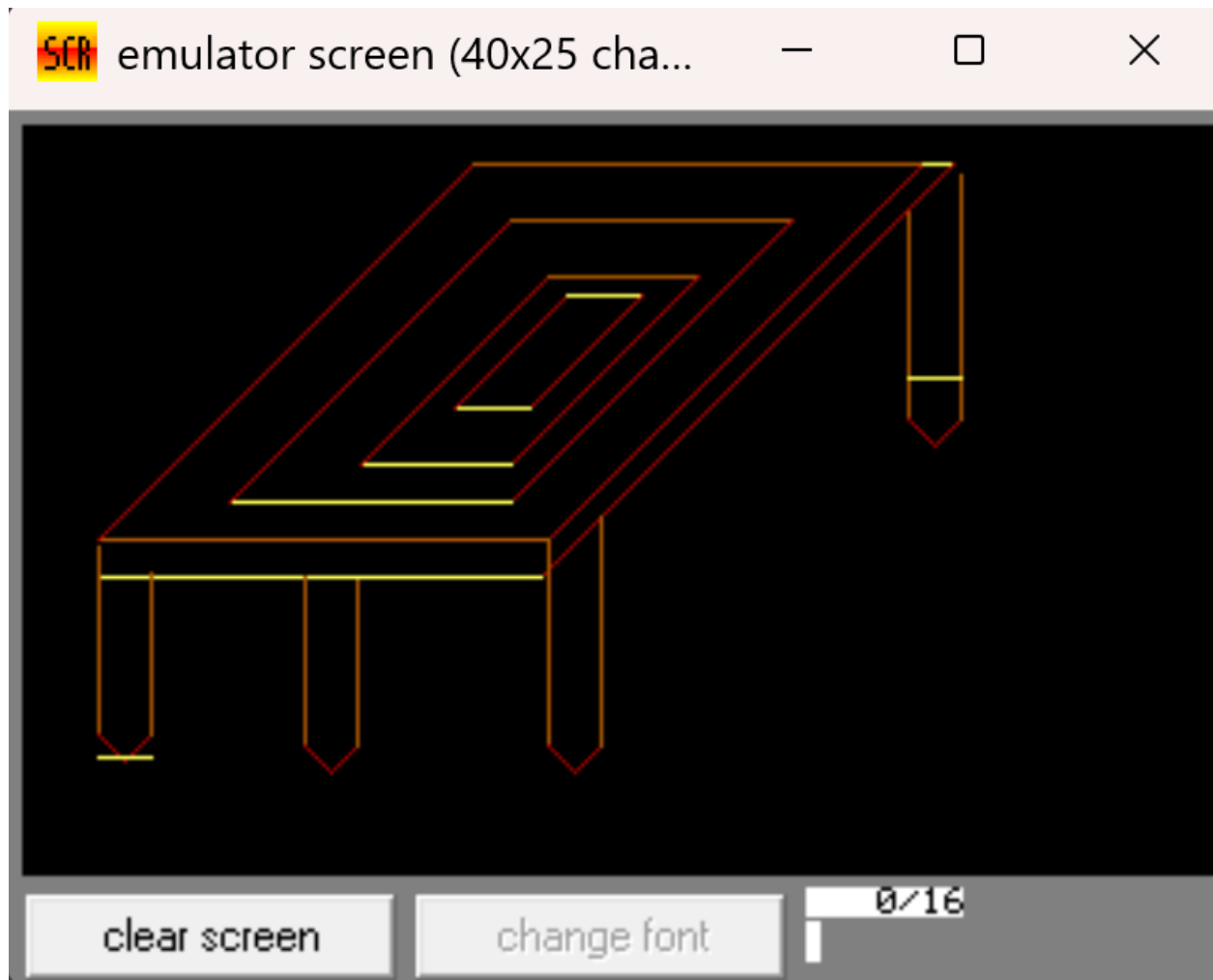
    MOV AH, 4CH

    INT 21H

MAIN ENDP

END MAIN
```

Output:



Code Description:

This Assembly program renders a "**Wooden Style**" Table in VGA Graphics Mode 13h (320x200 resolution).

- **Setup:** Initializes Mode 13h and defines a custom color palette (Brown, Red, Yellow) to simulate wood texture and shading.
- **Structure:**
 1. **Frame:** Draws a perspective parallelogram using loops to form the outer and inner table edges.
 2. **Legs:** Renders four vertical legs, using loops to calculate thickness and perspective.
 3. **Feet:** Adds distinct yellow bases to the bottom of each leg for detail.

4. **Pattern:** Draws three smaller concentric parallelograms in the center of the table surface to create a decorative top design.
- **Execution:** The program runs the drawing logic, waits for a user keypress, and then returns to text mode.